AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

22AIE498 PROJECT PHASE 1

FLOODSENSE: A MULTI-MODEL FRAMEWORK WITH RAG-LLM SYSTEM FOR AUTOMATED FLOOD REPORTING

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INTRODUCTION & MOTIVATION

- One of the most destructive natural calamities, floods seriously damage people's lives, property, and the environment.
- Manual inspection doesn't scale during large-scale disasters, but quick and accurate flood assessment is essential for efficient disaster response and resource allocation.
- There is a gap between detection and reliable, actionable insight since current AI methods either identify flood water without generating useful reports or generate reports without verifiable footing in actual facts.
- FloodSense creates an automatic flood assessment report from the input image. It then uses RAG and an LLM to cross-check past flood cases and conventional response protocols to recommend reaction steps.

LITERATURE REVIEW (VLM)

SNo	Title	Methodology	Drawbacks	Dataset information	Results
1.	**Integration of large vision language models for efficient post-disaster damage assessment and reporting (2026)**	In order to automate the entire post-disaster assessment, reporting, alarm creation, and resource allocation workflow, the article suggests DisasTeller, a multi-agent LVLM framework in which RAG works with four specialised AI agents.	Because of the framework's strong reliance on LVLM accuracy, errors in early-stage damage assessment could propagate to later agents, leading to incorrect resource allocation, reconstruction planning, and alarms, which would require human confirmation before deployment.	It evaluates the framework using established post-disaster imagery data from te Wajima City earthquake case study in Japan	DisasTeller finished all tasks in a matter of minutes after being validated on the Wajima City earthquake GPT-4o demonstrated noticeably higher accuracy than Gemma3, scoring 7.3/10 on disaster grading compared to Gemma3's 4.6/10.
2.	**Explainable flood damage assessment using multi-atrous self-attention and vision-language integration (2026)**	The suggested approach combines a LoRA-fine-tuned BLIP-2 vision-language model to produce disaster descriptions with a MASA-enhanced DeepLabV3+ segmentation model for multi-scale flood object segmentation.	End-to-end multimodal reasoning is limited and deployment complexity is increased by the framework's reliance on two independently trained models and distinct datasets, where the segmentation output is not immediately utilised by the vision-language model.	This study uses two datasets: (1) **FloodNet** (high-resolution UAV imagery for semantic segmentation including flooded buildings and roads) and (2) **LADI-v2** (10,000 low-altitude aerial images paired with descriptive textual annotations for vision-language captioning).	High segmentation accuracy for flood-affected structures was demonstrated by testing on the FloodNet dataset; the LoRA-fine-tuned BLIP-2 model produced context-aware disaster descriptions with an 80.77% BLEU score.

LITERATURE REVIEW (RAG)

SNo	Title	Methodology	Drawbacks	Dataset information	Results
3.	**CrisiSense-RAG: Crisis sensing multimodal retrieval-augmented generation for rapid disaster impact assessment (2026)**	The study suggests a split-pipeline architecture with distinct Text Analyst and Visual Analyst modules linked by a temporal-aware fusion engine that uses post-event images primarily for structural damage assessment and prioritises real-time textual evidence for estimating peak flood extent.	The framework's applicability to other disasters and geographical areas is questionable because it was only tested on Hurricane Harvey. It also depends on the quality and accessibility of other additional data sources. Prediction accuracy may suffer in areas with scant, noisy, or delayed data.	This study uses a curated Hurricane Harvey dataset comprising high-resolution aerial imagery tiles from NOAA, filtered tweets, emergency calls, and precipitation sensor data, all integrated for rapid impact assessment	The complete multimodal configuration reduced flood extent error by up to 8.62 percentage points compared to text-only inference, achieving flood extent error of 8.86–19.64% and damage severity error of 10.14–14.47% across four model backends when tested on Hurricane Harvey over 110 ZIP-code searches.
4.	**Training-free retrieval-augmented generation with reinforced reasoning for flood damage nowcasting (2026)**	uses labelled tabular flood records to propose a three-level reasoning-centric knowledge base. It predicts flood damage without task-specific LLM fine-tuning while producing explainable explanations by combining local neighbour retrieval, HUC12/global prototypes, reinforced reasoning, and a rule-based downgrading mechanism.	The framework's applicability to other areas and flood occurrences is not validated because it was only tested on Hurricane Harvey (Harris County, Texas). Additionally, it depends on the availability of past flood records because it uses a pre-built labelled knowledge base.	This study uses a curated dataset of 15,378 labeled 500m×500m grid cells in Harris County, Texas, for Hurricane Harvey, combining NFIP claims, FEMA Individual Assistance records, and 14 built-environment, topographic, hydrologic, and event-specific predictors for Property Damage Extent (PDE) nowcasting.	The framework was 35.9–81.9x more cost-effective than bigger LLM backbones, requiring no task-specific training, and reached 0.613–0.668 overall accuracy and 0.757–0.896 damage-class accuracy across seven LLM backbones, nearing the supervised baseline (0.714 accuracy).

LITERATURE REVIEW (RAG)

SNo	Title	Methodology	Drawbacks	Dataset Information	Results
5.	**Remote Sensing Retrieval-Augmented Generation: Bridging Remote Sensing Imagery and Comprehensive Knowledge with a Multi-Modal Dataset and Retrieval-Augmented Generation Model (2026)**	The authors provide RSWK (satellite images + structured knowledge) and RS-RAG, a RAG framework that encodes text and pictures into a shared vector space, retrieves knowledge through cross-modal similarity, and combines it with VLM prompts for VQA, categorisation, and captioning based on knowledge.	Scalability is hampered by computational overhead from retrieval and LLM fusion, generalisability is restricted by the fixed dataset, and task-specific parameters affect performance.	In his paper he proposes the Remote Sensing World Knowledge (RSWK) dataset, a large-scale multimodal benchmark consisting of 14,820 high-resolution satellite images of well-known landmarks in 16 categories from 184 countries, aligned with structured remote sensing domain knowledge and comprehensive world knowledge.	RS-RAG achieves significantly better performance than state-of-the-art baselines across all tasks with a BLEU-1 score of 0.490 (image captioning), 79% overall accuracy (image classification) and 0.685 CIDEr score (VQA). Ablation studies reveal that visual similarity weighting and retrieval granularity are task-dependent for optimal performance.

LITERATURE REVIEW (SEGMENTATION)

S No	Title	Methodology	Drawbacks	Dataset Information	Results
6.	**Multimodal Mamba with multitask learning for building flood damage assessment using synthetic aperture radar remote sensing imagery (2025)**	In order to concurrently estimate damage, flood extent, and footprints, the authors offer Flood-DamageSense, which combines SAR, VHR optical basemaps, and flood-risk layers via a Mamba-based semi-Siamese encoder with multi-task decoders. The results are aggregated to building-level maps via median pooling across polygons.	The generalisability of claim-based PDE ground truth is restricted to the Hurricane Harvey case study; it under-represents uninsured damage and introduces noise from floodwater loss mismatch, urban floodwater identification fails, and localisation deteriorates without VHR optical data.	This study uses the UrbanSARFloods dataset (Sentinel-1 SAR imagery from 18 global flood events including Hurricane Harvey in Harris County), pre-event VHR optical imagery, and building footprint data with ground truth PDE labels derived from NFIP and FEMA IA claims.	Flood-DamageSense substantially outperforms state-of-the-art baselines, with a mean F1 improvement of up to 19 percentage points, achieving a building-level harmonic mean F1 of 0.2119 (standard) and 0.4230 (upper-adjacent) for damage classification, with the historical flood-risk layer being the single largest contributor to performance improvement.

LITERATURE REVIEW (LLM)

SNo	Title	Methodology	Drawbacks	Dataset Information	Results
7.	**LLMArk: Instance-aware foundation model for flood risk assessment (2026)**	The suggested approach, an instance-aware multimodal large language model (MLLM) for flood risk assessment, combines a LoRA-fine-tuned vision-language model trained on the recently released VLFRA dataset with a YOLO-based Instance Perception Token.	Because the recommended method mostly depends on the proprietary VLFRA dataset and expert annotations that are personally verified, it is difficult to generalise to previously encountered flood events or geographies. Precise object detection is also essential to its efficacy; instances that are veiled or poorly identified can lower the standard of hazard assessment and rescue recommendations.	This paper used VLFRA which is a large-scale multimodal flood dataset that uses multimodal large language models to enable instance-aware flood risk assessment and rescue guidance. It consists of 50K flood-specific QA conversations, 12K annotated human and vehicle instances, and 5K real-world flood photos.	Using only a 7B-parameter model, LLMArk achieved more accurate instance-level flood risk assessment, object detection, and rescue guide generation than cutting-edge multimodal LLMs (such as ChatGPT-4o, Llama 3.2-Vision, and Qwen2-VL) across all 12 evaluation metrics.
8.	**DisasterM3: A Remote Sensing Vision-Language Dataset for Disaster Damage Assessment and Response (2026)**	presents DisasterM3, a multi-hazard, multi-sensor, multi-task remote sensing vision-language dataset that allows for VLM reasoning and report creation. It contains 26,988 bi-temporal optical/SAR images and 123k instruction pairs for nine disaster assessment and response tasks.	It is essentially a benchmark study and dataset rather than a new catastrophe assessment model, and current VLMs continue to perform poorly, particularly on SAR images and challenging reasoning tasks.	This paper introduces **DisasterM3**, a large-scale remote sensing vision-language dataset containing 26,988 bi-temporal images and instruction pairs covering 36 historical disaster events across 10 disaster types and 5 continents, with 9 disaster-related tasks.	Fine-tuning on DisasterM3 results in up to 10.4% improvement in QA accuracy, 2.1 points higher report scores, and 40.8% improvement in referring segmentation, with PSALM-1.3B achieving 50.5% cIoU (optical-optical) and GPT-4.1 achieving the best overall accuracy of 42.3% among zero-shot models.

RESEARCH GAP

- **Lack of robust multi-class flood segmentation:** The majority of flood detection techniques currently in use perform binary water/no-water segmentation using a single technique without benchmarking across multiple approaches (classical, foundation models like SAM, transformer-based segmenters) or differentiating between flood water, debris, and damaged structures. This results in coarse, unreliable masks that don't capture the full picture required for damage assessment.

- **Lack of proper report generation:** Current automated systems either use standalone LLMs to generate situational reports without connecting claims to verifiable image-derived features or authoritative sources, or they output raw segmentation masks without any interpretation. This results in reports that are either unusable by responders or prone to hallucinated, untraceable severity/impact assessments.

PROBLEM STATEMENT

"Current flood assessment systems are inadequate in two ways: text-based systems generate reports from secondary web sources without direct visual grounding, making them slow, incomplete, and unverifiable against actual site imagery, especially early in a disaster; segmentation-only models identify flooded regions but leave severity interpretation and reporting to humans. At the moment, no solution combines report production with fine-grained flood segmentation so that each claim may be linked to visual proof. "

Scope: FloodSense is an end-to-end demo dashboard that includes automated flood-water segmentation from photos (benchmarking five CV approaches), RAG-based historical and protocol data retrieval and structured severity reports created by LLM.

Boundaries: The project relies on existing imagery and knowledge-base documents rather than live field data collection. It does not include live drone data ingestion, or generalization to other disaster types beyond a proof-of-concept.

OBJECTIVES

- Go beyond binary flood detection by implementing multi-class flood/damage segmentation (differentiating water, structural damage) with a per-class F1-score.

- Using flood-labeled test images, benchmark five flood segmentation techniques (traditional thresholding, SAM, VLM-based, Grounding DINO+SAM2, OneFormer/SegFormer) and determine which strategy or approaches perform the best, aiming for an better IoU and Dice score.

- Create a RAG retrieval pipeline with a better retrieval precision for pertinent document retrieval over a knowledge base of historical flood records and disaster protocols.

- Create structured, grounded reports (impact estimation, severity assessment, and action suggestions) that be linked to segmentation output, verified by manual or expert review.

- Provide a working end-to-end demo dashboard (picture upload structured report) that is appropriate for lightweight field deployment and has an average inference-to-report duration per image.

METHODOLOGY



Classical Thresholding

SAM

Visual LLM

Grounding DINO+SAM 2

OneFormer/SegFormer

Segmented Image



Class 0: Building Flooded
Class 1: Building Non-Flooded
Class 2: Road Flooded
Class 3: Road Non-Flooded
Class 4: Water
Class 5: Tree
Class 6: Vehicle
Class 7: Grass

Preprocessing

LLM Generates Report

Previous Historical flood reports,Government disaster guidelines,Flood Safety SOPs(standard operating procedures),Infrastucture Guidelines from NDMA and other sources

Knowledge Base

Embedding and Indexing(ChromaDB)

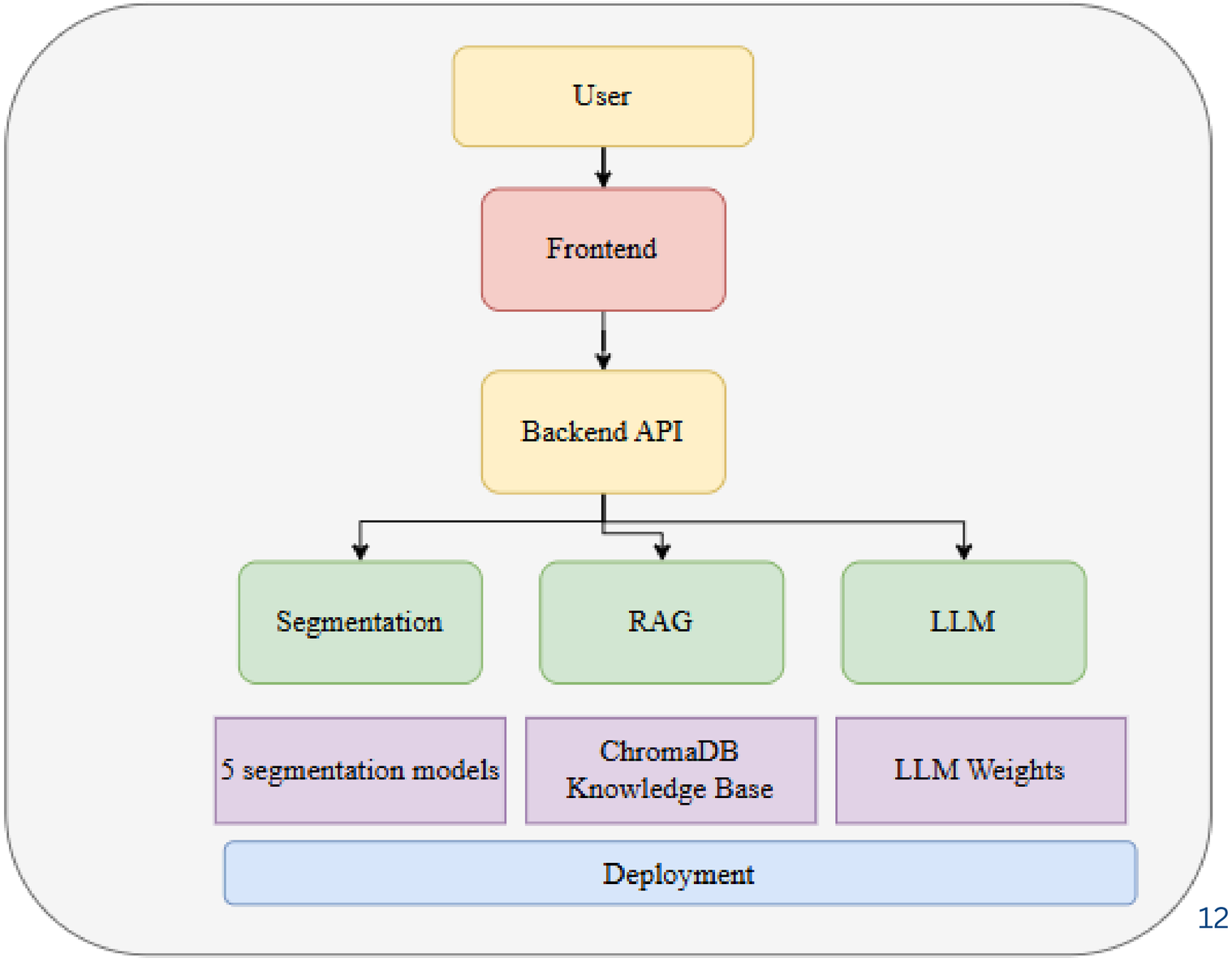
Retrieval Process

Retrieval Evaluation

a set of retrieved, relevant documents grounded in historical and procedural knowledge.

METHODOLOGY

ARCHITECTURE



TECHNIQUES AND TOOLS

Models Used:

- Classical thresholding - Traditional CV method for segmentation
- Segment Anything Model (SAM)
- Grounding DINO + SAM 2
- Transformer-based semantic segmentation
- Vision-Language Model (VLM)
- RAG retrieval

Datasets:

- FloodNet - Drone (UAV) images of urban flood scenes with pixel-level segmentation annotations.
- AlleyFloodNet - Developed by Ook Lee and Hanseon Joo, Department of Information Systems, Hanyang University, South Korea.
- Deep Flood - Developed for deep learning-based flood detection and depth estimation. Distributed through HydroShare as the Urban Flood Image Dataset.

TECHNIQUES AND TOOLS

Software / Frameworks:

- PyTorch (a fundamental deep learning framework)
- HuggingFace Transformers (OneFormer or SegFormer)
- SAM + Grounding DIN + OpenCV
- Sentence-Transformers (RAG embedding generation)
- ChromaDB (retrieval vector storage)
- LlamaIndex (RAG arrangement)
- React (Dashboard)
- Docker (deployment in containers)

TECHNIQUES AND TOOLS

Hardware:

- Local Development: NVIDIA GTX for lightweight models and debugging.
- Cloud computing: Kaggle GPUs for SAM, SAM 2 plus Grounding DINO, and VLM inference; Google Colab (T4 16GB).
- The system needs at least 50GB of storage and 16GB of RAM.
- Storage: For durability, the Chroma vector database, model checkpoints, and datasets are stored on Google Drive.

Platform:

- Dashboard created with React (upload image → report generation)
- RAG pipeline + CV models through a backend API (Flask)
- A small, deployable local or cloud-hosted demo

EXPECTED OUTCOMES

Prototype / Deliverables:

- Complete working demonstration: uploading an image, segmenting it, and creating a structured report on a simple dashboard
- Retrieving pertinent historical and protocol papers using a functional RAG pipeline
- Five segmentation techniques are compared using flood imagery.
- Structured reports produced by LLM (severity rating, impact assessment, suggested reaction actions) with traceable reference to retrieved documents or CV output

Performance Metrics:

- Segmentation - IoU, Dice Score, Accuracy, Precision and Recall
- RAG - Precision@K

INDUSTRY RELEVENCE

Use Cases:

- Quick evaluation of catastrophe damage for emergency response teams during floods
- Automated situation reporting for government disaster management authorities, NGOs, and humanitarian agencies
- Support for resource allocation: giving priority to rescue and relief efforts in high-severity areas

Opportunities for Collaboration:

- For data access and protocol validation, **government disaster management** organizations (State/National Disaster Management Authorities)
- NGOs and relief agencies for field validation and feedback loops, such as the Red Cross and local disaster response units

TIMELINE

Phase 1: till Aug 25th

Multi-Class Segmentation

- Using five CV techniques (thresholding, SAM, VLM, GroundingDINO+SAM2, SegFormer/OneFormer) to identify and categorize floodwater, debris, and impacted structures.
- Using IoU, Dice, precision/recall, and inference time, choose the model or models that perform the best.

Phase 2: Aug 25th - Sep 18th

LLM + RAG

- Create a retrieval pipeline for historical flood and awareness records using ChromaDB and sentence transformers.
- Combine recovered context and CV-extracted characteristics to create organized, severity reports using LLM

Phase 3: Sep 19th - Oct 9th

UI & Deploy

- Create a lightweight dashboard for uploading images and displaying structured reports using React.
- Package and deploy the complete pipeline (backend API + models).

TEAM ROLES

Varshitha Thilak Kumar — Computer Vision (Classical + Promptable Segmentation)

- Establish and adjust the baseline for classical thresholding
- Put SAM and SAM 2 + Grounding DINO pipelines into action.
- Get datasets ready and preprocessed for these techniques.
- Participate in CV benchmarking for their models (IoU, Dice, precision/recall, inference time).

Shreya Arun— RAG & Knowledge Base

- Create a knowledge base that includes historical flood records and disaster response procedures.
- Configure the vector store (Chroma) and embeddings (sentence-transformers).
- Construct and adjust the retrieval pipeline; assess the recall and precision of the retrieval
- Take responsibility for the groundedness assessment process.

Siri Sanjana S— Computer Vision (Transformer + VLM Segmentation)

- Use a segmenter based on transformers (OneFormer/SegFormer)
- Put in place a segmentation process based on VLM
- Manage the extension of multi-class segmentation (water, debris, structural damage, vegetation)
- Participate in their models' CV benchmarking and compile the complete 5-method comparison report.

Anagha Menon — LLM Integration & Platform/Deployment

- Build the report-generation LLM pipeline (prompt design, structured output)
- Integrate CV outputs + RAG outputs into the LLM synthesis stage
- Build the Flask backend and Streamlit/Gradio frontend
- Handle deployment (Hugging Face Spaces/cloud VM) and end-to-end pipeline integration

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DATASET LINK

1. FloodNet - Drone (UAV) images of urban flood scenes with pixel-level segmentation annotations. - https://github.com/BinaLab/FloodNet-Supervised_v1.0
2. AlleyFloodNet - Developed by Ook Lee and Hanseon Joo, Department of Information Systems, Hanyang University, South Korea. - https://www.kaggle.com/datasets/seonyseony/alleyfloodnet
3. Deep Flood - Developed for deep learning-based flood detection and depth estimation. Distributed through HydroShare as the Urban Flood Image Dataset.- https://www.hydroshare.org/resource/24866122a6ee456c8f7c80aa87a9abcb/

THANK YOU